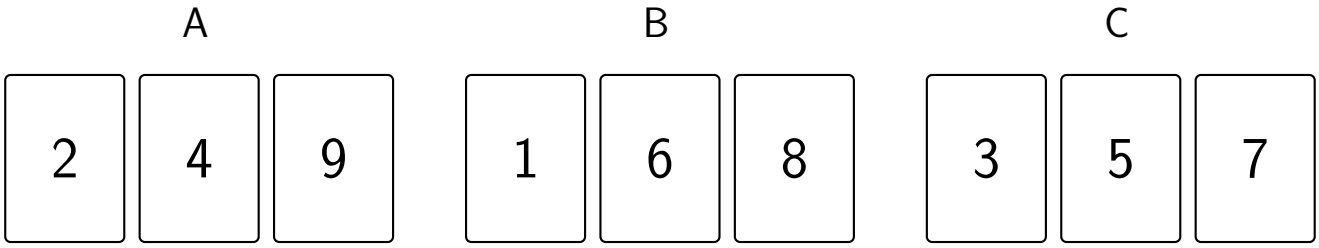
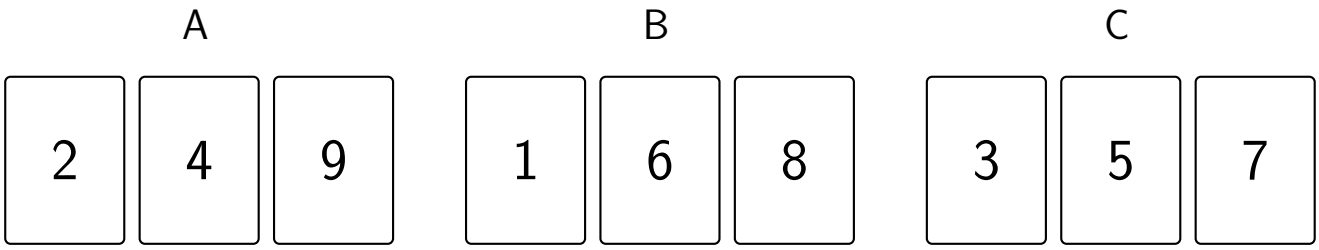
Decks A, B and C are bags of three number cards, shown below. Mix each bag on its own. In a game, each player takes one deck. Each round, draw one card from each player's bag without looking; the bigger number wins the round. Put both cards back before the next round.

Problem 1: One player picks one of the decks A, B, and C. Then the other player picks a different deck. They play ten rounds while a third player draws the cards and keeps score. Rotate jobs and play again. Is it better to pick first or second?

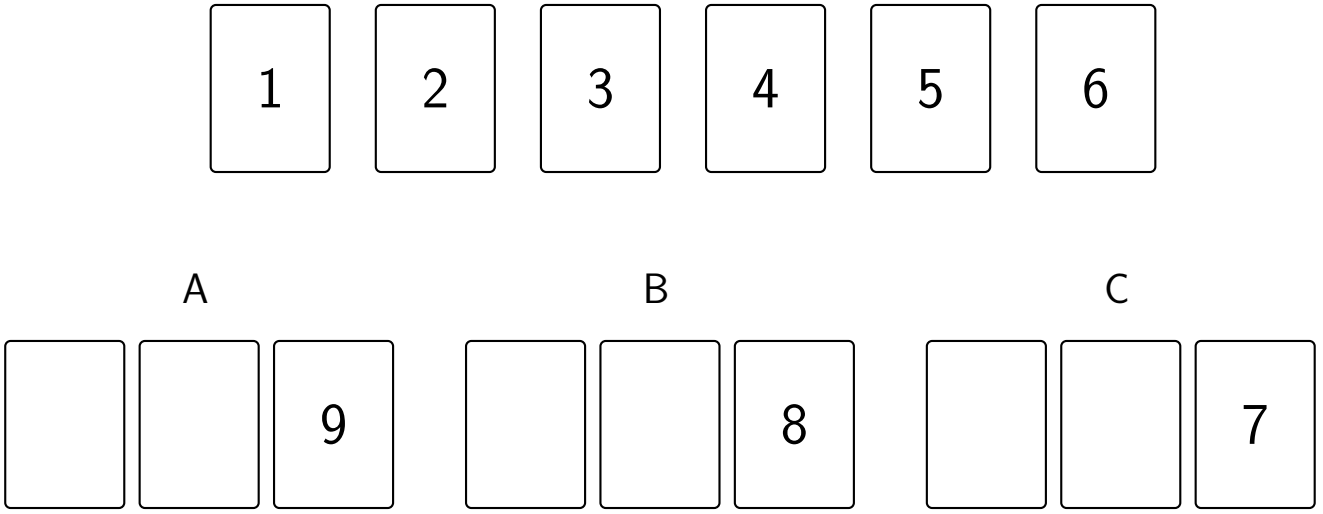


	First pick	Second pick
example	A, 4 wins	C, 6 wins

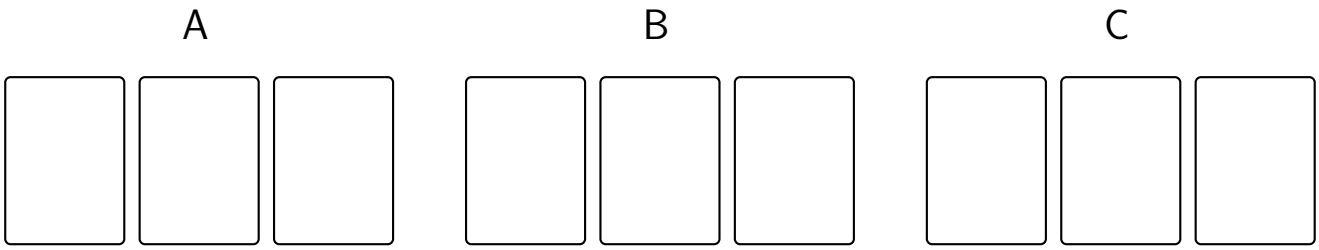
Problem 2: For each pair of decks, find how many of the possible card pairs each deck wins. Is there a deck that wins more than half its pairs against both others? Do the three deck totals settle this question?



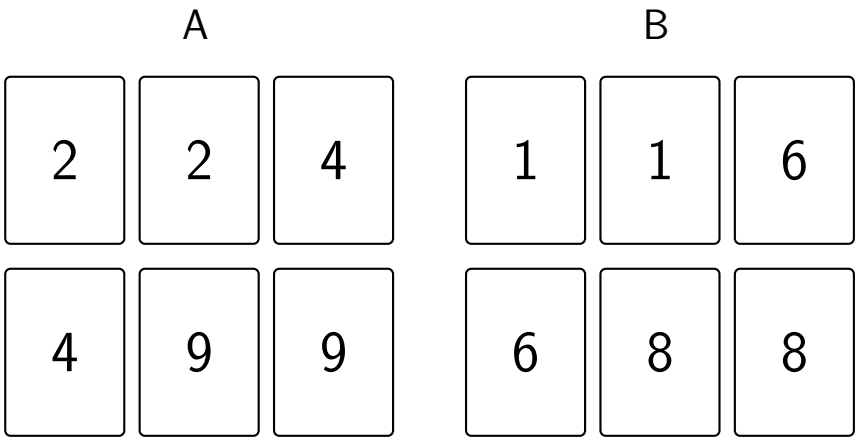
Problem 3: Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

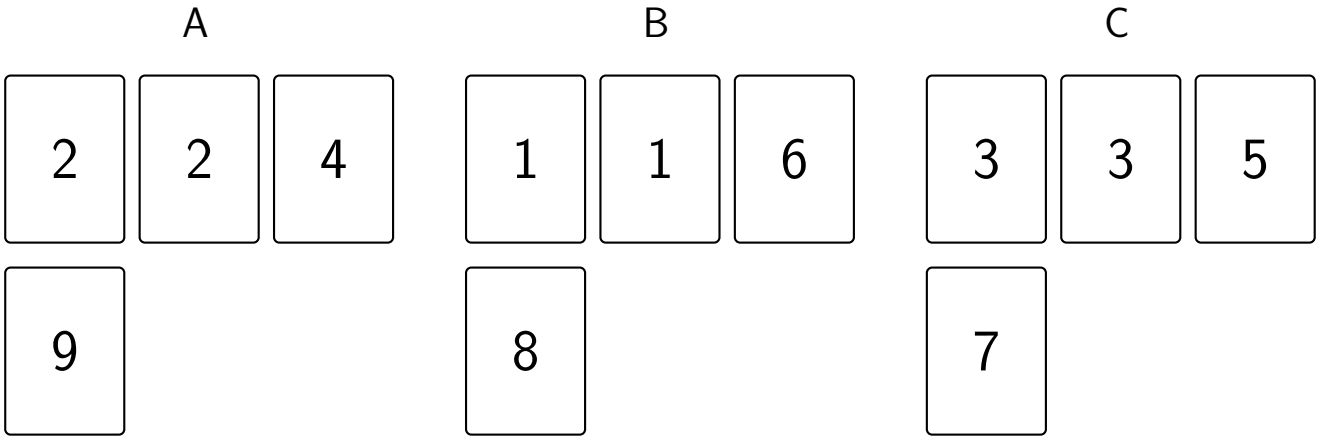


Problem 4: Write numbers from 1 to 6 on these nine cards. A number may repeat within one deck, but it cannot appear in two different decks. Make A win more than half its pairs against B, B against C, and C against A.

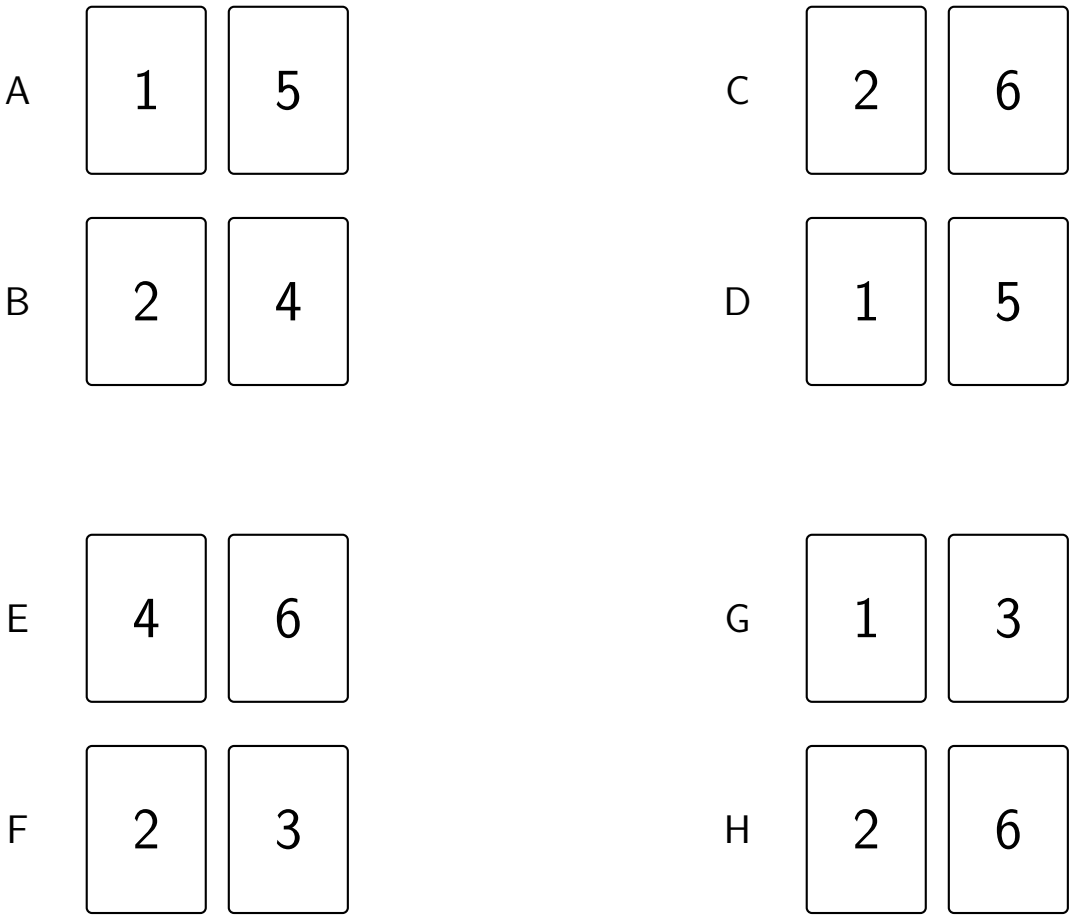


Problem 5: Find A's chance of beating B with the six-card decks. Then find each deck's chance of winning against each other deck with the four-card decks. Which chances changed from Problem 2?

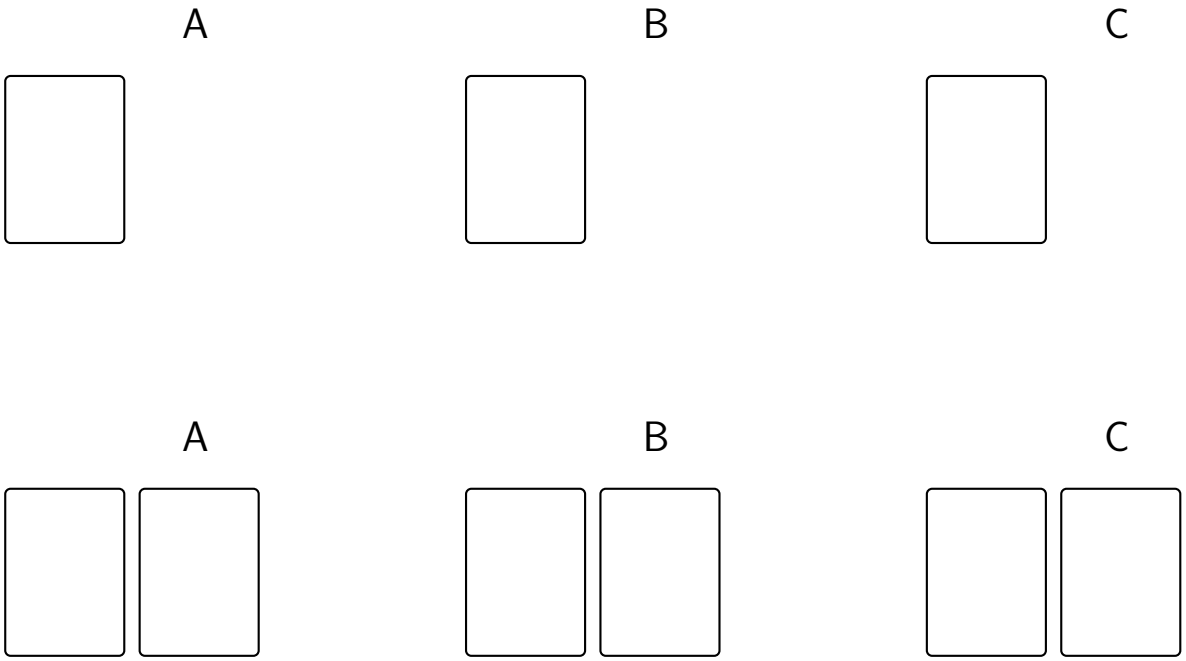




Problem 6: Decide which deck, if either, wins more than half the pairs in each comparison. Find a rule that works for every pair of two-card decks with no number shared between decks, and explain why it works.



Problem 7: Can three one-card decks make A win more than half its pairs against B, B against C, and C against A? Can three two-card decks do it? Explain for any choice of numbers. What is the fewest cards per deck that allows this, with all decks the same size?



Problem 8: Use cards 1 to 9 once each in three decks of three. Can A win at least six of its nine pairs against B, B at least six against C, and C at least six against A? Find an arrangement, or explain why it is impossible.

